

Questions-Please Submit Short Answers to These Questions

- 1) What is the most common event generator used to describe tree-level QCD processes and jets? *Tree level means Leading Order*
- 2) What is the name of the hadronization model is used for the answer to 1)?
- 3) What are the two main kinds of jet finding algorithms?
- ~~4) Which of the two answers in 3) is not actually used anymore (because it's not "infrared safe" e.g.)?~~
- 5) Which jet finding algorithm is almost exclusively used by CMS and ATLAS at the LHC?
- ~~6) What single letter capitalized variable is used to represent the angular size of jets and jet subcomponents, and what formula defines it? (in terms of phi angle differences and what other quantity)~~
- ~~7) [This is the ONLY question on the whole final assignment NOT from this reading, but for which the answer can be easily found in the rapidity slides]: what are the units of the answer to 6), and why would/could the "other quantity" besides phi angle mentionned in question 6 and thus the answer to 7 itself be also measured in this unit] Be brief in the answer ask me if you can't state it briefly.~~

Questions (continued)

- Now on the Fragmentation Function section:
- 8) What variable is the argument of the Fragmentation Function $D()$ and what is its physical significance meaning what does its value (the argument variable) represent.
- ~~9) In one of the FF readings, an X with a circle around it represents INTEGRAL CONVOLUTION (not tensor product)...what three mathematical functions/quantities are generally convoluted in a factorization calculation? Which of them is not included for e^+e^- annihilation factorization calculation and why?~~
- ~~10) According to DELPHI measurements, for a bunch of jets all of the same momentum, which type, gluon or quark jets, will have particles with on average higher momentum?~~

More Questions... continued-String Model

Concerning the readings in in Wong -Schwinger Mech 5.1 & 7.1)
optionail warmup: Is equation 5.2 ~the same as or not the same as
the case of an infinite plane of charge in E&M? (no more than 1
sentence explanation can be optionally included).

- 11) What is the order of magnitude of the string tension constant κ on page 63 of wong
- 12) What is the difference btw Eq.'s 5.4 and 5.5 and what physical phenomenon causes it
- 13) ~~Wong says that there is a diagram like in Figure 7.1a /b for each value of W_{H1} that the exchanged or virtually produced (aka s-channel) particle possesses.~~
- 14) A Regge trajectory is a _____ (what kind of "graphical") relationship between the answer to 13) and _____ (what?)
- ~~15) Name one kind of experimental data Wong says the string model cannot explain well.~~
- 16) In Figure 7.4 a) what are the values of the slopes of the line segments and is the meson the diagram is supposed to represent at rest or moving?
- ~~17) Describe what oscillation of the string which the "yo yo model" describes and what quantity/property of the meson the string represents can the energy associated with this oscillation be related to?~~
- ~~18) What kind of geometric curve (a conic section) does the Lund model propose that all particle production points in spacetime fall along approximately? Which part of the graph/trajectory traced out on figure 7.6 upper or lower part?~~

Even More Questions – DPM Pomeron

- Remember also my slides interspersed with the external sources in the pdf file are also part of the required reading assignments. This sections concerns more about the Pomeron and the Dual Parton Model Section from Wong reading
- 19) What kind of hadron is likely to be produced in a inelastic (but “pseudo-elastic” ie proton intact) diffractive event/interaction?
- ~~• 20) What perturbative model describes the interaction of a virtual photon splitting into a color/anticolor charge pair which interacts with the hadron? Hint the slides say Color Transparency, a somewhat well known but also perhaps somewhat maligned theory sometimes used to describe lots of Jlab data, is an offshoot of this model?~~
- ~~• 21) What two variables is the Veneziano scattering amplitude expression (Eq 8.1) invariant to the exchange of, reminiscent of the exchanges/permutations of different Feynman diagram results for various kinds of “crossing” relations in the cross section calculations + mostly pointed out in relation to Halzen and Martin results?~~
- 22) What kind of relationship does the reading in Wong’s D.P.M section ascribe between the total cross section and the scattering center of mass energy?
- 23) Wong says the Pomeron has the quantum numbers of the _____ (what)?

Final group of Questions – Bag Model

- ~~24) List some variables/quantities whose values are important in the Bag model.~~
- 25) What does the constant B represent in the Bag model?
- 26) How does the $1/4$ power arise in relation to the any of those expressions for B ? (ie in the derivation of those equations, relating what to what or finding what..etc. what part of those derivations)
- ~~27) Name two physical quantities (any of the answers to 24 are OK) are derived in the reading selections/slides using the Bag Model and what are their values?~~
- 28) What is the critical temperature for QGP (quark gluon plasma) formation Wong gives from Bag model estimates?
 - The real temp found at RHIC was about 170-180 MeV